



COMP 163 – Discrete Structures: 001

Spring 2026

Tues/Thurs 08:30 AM - 09:45 AM

Instructor – TaiNing Wang

Office Hours - Friday 2-3 PM (tentative) (Location: Doyle 212) & by Appointment.

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This policies laid out in this syllabus are subject to change due to unforeseen circumstances, scheduling conflicts, etc.

Course Description: This course covers the mathematical foundations of computer science, including such topics as set theory, propositional logic, induction and other proof techniques, functions and relations, combinatorics, probability, and algorithm analysis.

A major goal of this course is to develop the mathematical maturity needed for further study of advanced topics in theoretical computer science. Thus, proof-writing and communication skills are strongly emphasized.

Textbook: Johnsonbaugh, Richard: *Discrete Mathematics*, 8th Edition

Please note that the textbook is not required explicitly (i.e. homeworks and other class exercises will not rely on the student purchasing the textbook). However, it is recommended that students use the textbook as a guide to enhance the understanding.

Prerequisites: Math 117 (Precalculus I) or equivalent placement

Familiarity with concepts from high-school Algebra I, II and Pre-calculus is necessary. Exposure to proof-techniques such as those taught in high-school Geometry is also helpful. The course does not require any programming experience.

Learning Outcomes:

1. Reason about abstract sets, and perform basic set-theoretic operations
2. Demonstrate facility with propositional logic, quantification of logical statements, and basic laws of inference
3. Construct rigorous proofs of basic mathematical statements including direct-proof, contradiction, and proof by induction

4. Demonstrate facility with binary, hexadecimal, and floating-point arithmetic
5. Use combinatorial methods to enumerate discrete outcomes
6. Use of basic probability axioms, including conditional outcomes to assign probability measures to discrete outcomes
7. For COMP-400E students: All of the above with a deeper understanding

Homework:

There are four homework assignments, each worth 30 points. The assignments are posted on Gradescope and SAKAI, and completed homework must be uploaded/scanned to Gradescope by the specified due date in **pdf format**. **Paper homework is not accepted**. If a student relies on photographs of written work to document solutions, it is incumbent upon the student to ensure that the work is visible for the grader. Illegible and/or disorganized homework solutions may not be graded. It is recommended that students typeset (e.g. $\text{\LaTeX}$) their assignment submissions.

Homeworks consist of two parts. Part I consists of simpler problems, and are graded primarily for completeness. Part II consists of more difficult exercises; some of the questions in Part II will be graded in detail. The graded items are not known in advance of the submission date, thus it is recommended that students attempt all exercises. 10 pts will be awarded for general completeness of the assignment, while the graded problems will account for the remaining 20 pts.

Solutions to all homework exercises will be distributed after the due date on Sakai. Any perceived errors in grading should be resolved by submitting a regrade request on Gradescope. While the instructional team strives to minimize errors in grading, errors do happen, so please review all graded work.

You are encouraged to work together with your classmates on the homework assignments! However, everyone must submit individual assignment solutions. Finally, without a legitimate excuse, **late homework will not be accepted**. The due dates for the homework are in the last column of the table found at the end of the syllabus.

Typesetting in $\text{\LaTeX}$:

Proficiency in mathematical typesetting is an extremely valuable skill. The following links are helpful in getting started with $\text{\LaTeX}$:

1. https://www.overleaf.com/learn/latex/Learn_LaTeX_in_30_minutes
2. https://www.reed.edu/academic_support/pdfs/qskills/latexcheatsheet.pdf
3. https://math.hws.edu/gassert/LaTeX_Guide_Title.pdf

Quizzes:

There will be 6 in-class quizzes throughout the semester (refer to the schedule for specific dates). On quiz days, the class will meet online. For each quiz, you will have a set amount of time to complete it. If you submit your answer on time, you will earn 5 points, regardless of whether your answer is correct or not. After the quiz, the instructor will ask for volunteer(s) to present their answers by verbally explaining their solutions to the class. If you present for the quiz, you will earn an additional 3 points, also regardless of the correctness of your answer.

At the end of the semester, your best 4 quiz scores will count toward your final grade. The maximum score for the quizzes is 20 points. If your total quiz score exceeds 20, you can earn up to 3 extra credit points for the course. There will be no make-up for the quizzes.

For example, suppose Student A earns the following scores across the 6 quizzes: 5+3 points on Quiz 1 (submission and presentation), 5 points on Quiz 2 (submission), 5+3 points on Quiz 3 (submission and presentation), 0 points on Quiz 4 (no submission), 5 points on Quiz 5 (submission), and 5 points on Quiz 6 (submission). The best 4 quiz scores are 5+3, 5+3, 5, and 5, adding up to 26 points. Since the maximum score for the quizzes is 20 points, Student A will receive 20 points for the quizzes and 3 points as extra credit for the course.

Exams:

There are two in-class exams. Both exams are **closed notes and closed book, no calculators, and some form of ID will be required**. Note that **attempts to seek any sort of outside help** (internet, other individuals, etc) during exams constitutes academic misconduct, and will be dealt with accordingly. Exams may only be missed under emergency circumstances (e.g. illness), and provable documentation of the circumstance is required. Note that work/other course commitments **do not** constitute a valid reason to miss an exam. If an exam is missed (with a valid reason), either a makeup-exam must be coordinated with the instructor (if available), or the weighting of the other grading items will be adjusted to account for the missed exam.

If an extraordinary circumstance arises that prevents a student from taking the Exam II, the student must fill out the following Incomplete Form, and send the completed form to the instructor within 72 hours of the exam date (with appropriate supporting documentation). If the form is not submitted to the instructor by email within 72 hours, a score of 0 will be entered for Exam II when calculating the final course grade.

https://www.luc.edu/media/lucedu/registrationrecords/pdfs/UGRD_Request_%20for_Incomplete%20-%20August272024.pdf

Attendance:

Attendance during lectures is highly recommended, and regular participation is correlated with higher grades in the course. Attendance is required for all examinations.

The university's statement about the observation of religious holidays is available here:

<https://www.luc.edu/campusministry/faithprograms/interfaith/religiousobservances/>

Piazza:

The piazza forum allows students to ask questions about homework and class material. The instructor will moderate the forum and answer questions as able, however it is also recommended that students help their fellow classmates by answering forum questions.

Grading:

The course grade is computed with the following weights:

Participation	5 %
Quizzes	20 %
Homework	20 %
Exam I	20 %
Exam II	30 %
Final Exam Activity	5 %

A portion of the final score is assigned to the instructor's assessment of student participation in the course. Participation includes asking/answering questions during lectures, offering to solve example problems, asking questions during office hours, and asking/answering questions on the piazza forum.

The following table reflects **guaranteed** letter grade cutoffs, and the cutoffs may be adjusted downwards as deemed appropriate.

≥ 90	A	80 - 82	B	70 - 72	C	60 - 62	D
87 - 89	A-	77 - 79	B-	67 - 69	C-	≤ 59	F
83 - 86	B+	73 - 76	C+	63 - 66	D+	-	-

COMP-400E:

Graduate/Post-Baccalaureate students enrolled in COMP-400E are expected to perform at a higher standard than those students enrolled in the corresponding undergraduate section. Those students are required to complete more challenging homework/exam problems and be subject to stricter grading standards commensurate with the performance that is expected of a graduate student.

Office Hours:

Please reserve office hours for questions relating to homework/course material. Questions regarding grading should be resolved via the Gradescope regrade process. Office hour times may be modified to accommodate the need for exam reviews, and updates will be made via email.

Incomplete Grades:

Note that Incomplete grades (I) are only assigned in extraordinary circumstances. It is the student's responsibility to first discuss the matter with his/her Academic Advisor before requesting that an Incomplete be granted. If the Academic Advisor believes that circumstances warrant an Incomplete, then the student may request one from the Instructor.

Accommodations:

Accommodations for coursework must be approved by the Student Accessibility Center (SAC), and a Letter of Accommodation (LOA) must be provided to the Instructor at the start of the semester. Requests for accommodations will not be granted without a LOA.

Academic Integrity:

Academic integrity is taken very seriously. If you consult any additional resources other than the textbook and the course materials for Homework, citation of all resources consulted is required.

Note that using ChatGPT or other generative AI to generate text for Homework or Exam is strictly forbidden. Turnitin (turnitin.com) has capability of AI writing and ChatGPT detection, and may be used on some of the submitted works in this course.

In order to avoid the appearance of academic plagiarism, **always cite all resources consulted** (outside of the course materials). The first instance of cheating results in a zero on the assignment or exam, and the second instance results in failure of the class. You can read more at

<http://www.luc.edu/academics/catalog/undergrad/reg>

Academic Calendar:

Refer to the following page for information about registration, deadlines to drop the course, and other administrative dates.

Information about Title IX:

As an instructor, I am a Responsible Campus Partner (“RCP”) under Loyola’s Comprehensive Policy and Procedures for Addressing Discrimination, Sexual Misconduct, and Retaliation (available at luc.edu/equity). While my goal is for you to be able to engage fully and authentically with our course material through class discussions and written work, I also want to be transparent that as a RCP I am required to report certain disclosures of sexual misconduct (including sexual assault, sexual harassment, intimate partner and/or domestic violence, and/or stalking) to the Office for Equity & Compliance (“OEC”). As the University’s Title IX office, the OEC coordinates the University’s response to reports and complaints of sexual misconduct (as well as discrimination of any kind) to ensure students’ rights are protected.

As an instructor, I also have an obligation under Illinois law to report disclosures of or suspected instances of child abuse or neglect (<https://luc.edu/hr/legal-notices/mandatedreportingofchildabuseandneglect>).

The University maintains such reporting requirements to ensure that any student who experiences sexual/gender-based violence receives accurate information about available resources and support. Such reports **will not generate a report to law enforcement** (no student will ever be forced to file a report with the police). Additionally, the University’s resources and supports are available to all students even if a student chooses that they do not want any other action taken. If you have any questions about this policy, you are encouraged to contact the OEC at equity@luc.edu or 773-508-7766.

If you ever wish to speak with a **confidential** resource regarding gender-based violence, I encourage you to call The Line at 773-494-3810. The Line is staffed by confidential advocates from 8:30am-5pm M-F and 24 hours on the weekend when school is in session. Advocates can provide support, talk through your options (medical, legal, LUC reporting, safety planning, etc.), and connect you with resources as needed – without generating a report or record with the OEC. More information about The Line can be found at luc.edu/wellness.

Schedule:

The following table outlines the planned/tentative schedule for the semester. Holidays are identified in *italics*, while the exam dates are **bolded**. The fourth column has sections of the textbook (Johnsonbaugh unless otherwise noted) recommended to supplement understanding of the material from lecture. Finally, homework due dates and quiz dates are identified in the fifth column.

Week	Date	Topic	Readings	HW Due Dates (tentative)
1	Jan. 13	Course Introduction, Sets	1.1, 1.2	-
	Jan. 15	Propositional Logic (I)	1.3, 1.4	-
2	Jan. 20	Propositional Logic (II) <i>[Online]</i>	1.3, 1.4	Quiz 1
	Jan. 22	Quantifiers	1.5	-
3	Jan. 27	Logical Inference Rules	1.5	-
	Jan. 29	Boolean Algebra (I)	11.1, 11.2	HW1 / Jan. 30
4	Feb. 3	Boolean Algebra (II) <i>[Online]</i>	11.1, 11.2	Quiz 2
	Feb. 5	Combinatorial Circuits	11.3	-
5	Feb. 10	Methods of Proof (I)	2.1, 2.2	-
	Feb. 12	Review Tutorial	-	-
6	Feb. 17	Methods of Proof (II) <i>[Online]</i>	2.1, 2.2	Quiz 3
	Feb. 19	Exam I (08:30AM-09:45AM)	Week 1–5	-
7	Feb. 24	Methods of Proof (III)	2.1, 2.2	-
	Feb. 26	Proof by Induction (I)	2.4	HW2 / Feb. 27
8	Mar. 3	<i>Spring Break: No Class</i>	-	-
	Mar. 5	<i>Spring Break: No Class</i>	-	-
9	Mar. 10	Proof by Induction (II) <i>[Online]</i>	2.4	Quiz 4
	Mar. 12	Relations (I)	3.3, 3.4	-
10	Mar. 17	Relations (II)	3.3, 3.4	-
	Mar. 19	Functions (I)	3.1	HW3 / Mar. 20
11	Mar. 24	Functions (II) <i>[Online]</i>	3.1	Quiz 5
	Mar. 26	Introduction to Combinatorics (I)	6.1	-
12	Mar. 31	Introduction to Combinatorics (II)	6.1, 6.2	-
	Apr. 2	Introduction to Combinatorics (III)	6.3	-
13	Apr. 7	Numbering (I) <i>[Online]</i>	5.2	Quiz 6
	Apr. 9	Numbering (II)	5.2	HW4 / Apr. 10
14	Apr. 14	Probability Theory	6.5, 6.6	
	Apr. 16	Review Tutorial	6.5, 6.6	-
15	Apr. 21	Algorithm Time Complexity Analysis	4.2, 4.3	-
	Apr. 23	Exam II (08:30AM-09:45AM)	Week 6–14	-
16	May 2	Final Exam Activity (9:00AM-11:00AM)	-	Online